

Headline verdict

Mostly pass · 3 triangulated discrepancies · K⁺ alternative still open · cross-tool coverage closed

1SAR is a real and substantial improvement over the deposited starting model.

Modelling decisions on structure, geometry, ion identity (Mg²⁺ excluded), water, disulfides — all defensible.

Closed across the 04-30 / 05-01 / 05-04 re-issues

✓ Mg²⁺ alternative ruled out — CheckMyMetal-style IZI 16.35

✓ Ligand / metal density support confirmed — RSCC ≥ 0.96

✓ T13 data-quality cross-tool gap closed — truncate (NEW 05-04)

Still discrepant (3-tool confirmed)

x R-free reproducibly ≈ 0.21, agent reports 0.199

x Round 6 R-free 0.207 < round 7 R-free 0.213

x Asn A 39 is a Ramachandran outlier

x Density-peak inventory incomplete (23+ / 8– vs 9 listed)

Open question

K⁺ alternative for both ions

K⁺ has the best raw bond-length fit (IZI 0.13–0.44) but coordination number is too low for K⁺ canonical, and the protein is a known Ca²⁺ enzyme. Needs crystallisation buffer information.

Net acceptance against agent’s own success criteria — 3 / 6 pass cleanly

R-free improvement

✓ pass

R-free gap < 0.05

⚠ oracle fails

Ramachandran < 1%

✓ headline wrong

Clashscore < 5

✓ pass

No Δρ > 4σ

x fails

Data quality (T13)

✓ pass (NEW)